

SECTION 4 – SITE ENVIRONMENT

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4-1 INTRODUCTION

The rating and duty cycles of all subsystems are applicable only if the room environment is maintained as specified in the following sections. The environment must be constantly maintained (i.e. holidays, weekends, etc.) to prevent exceeding these restrictions. Subjecting the equipment to consistent excessive temperatures and humidity above specifications may shorten the life of the internal electrical components.

4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS**Note**

If specifications in this section are not strictly adhered to, failures of the ACGD modules in the ACGD/PDU Cabinet may occur.

Use the specifications listed in Table 4-1 for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmental controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

To help prevent a patient from feeling uncomfortably warm during a scan, ensure that the magnet room temperature does not exceed 70°F (21.1°C) maximum.

TABLE 4-1
TEMPERATURE AND HUMIDITY SPECIFICATIONS FOR SYSTEM WITH ACGD/PDU CABINET

AREA	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	HUMIDITY (%)	HUMIDITY CHANGE (%/Hr)	MAX. ROOM GRADIENT °F (°C)
Equipment Room at Inlet to Equipment	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)**
Magnet Room	59–69.8 (15–21)	5 (3)	30–60*	5	5 (3)
Operator's Control Room	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)
Note * Non-condensing humidity with 50% nominal at 65°F (18.3°C). ** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets.					

4-3 COOLING REQUIREMENTS

4-3-1 Air Cooling

Table 4-2 contains the heat output of the upgrade equipment listed in the typical site location. These values do not include people, lights, magnet equipment cabinets, and non-MR equipment. Actual site average values will vary depending on system use (ie. protocols used, patient load, etc.). Note any variations of equipment location for your site when calculating your cooling requirements for each room.

Requirements

Due to the large variation in Equipment Room heat loads, the air conditioner compressor should be equipped with unloaders or hot gas bypass to prevent moisture stripping of the evaporator coils.

The Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Enclosure Rear Pedestal, location of coil/bore cooling fans intake.

Recommendations

A dedicated air conditioner with a dual compressor is preferred to avert shutdowns during repair of the primary air conditioner.

It is recommended that a temperature and humidity recorder be used during preinstallation and during actual installation and placed near the Gradient Cabinet air inlets to establish the true criteria. Refer to cooling table calculator in this section for each room's cooling requirements.

GE recommends the use of a 12 inch high raised flooring system for the equipment room (10 inch minimum clearance from floor slab to underside of access flooring). Care must be taken in locating the air conditioning supply vents in the floor. The air conditioning supply vents should be located directly in front of the cabinet inlets.

TABLE 4-2
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	SIGNA HORIZON 1.5T MAX HEAT OUTPUT†						SIGNA HORIZON 1.0T MAX HEAT OUTPUT†					
	MAGNET ROOM (SEE NOTE 1)		EQUIPMENT ROOM		OPERATOR/ CONTROL AREA		MAGNET ROOM (SEE NOTE 1)		EQUIPMENT ROOM		OPERATOR/ CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
RF/Gradient Body Coil Assembly (BRM or CRM), Shim Coils	14,000	4100					10,925	3200				
Blower Box	3415	1000					3415	1000				
SRF Cabinet			8530	2500					8530	2500		
ACGD/PDU Cabinet			34,120	10,000					34,120	10,000		
System Cabinet			6140	1800					6140	1800		
Water Chiller for Gradient Coil			14,000	4100					14,000	4100		
MNS Cabinet*			2730	800			Not Applicable					
Shield Cooler Compressor			See Note 2						See Note 2			
Operator Workspace (See Note 3) with LCD Color Monitor					4095	1200					4095	1200
Main Disconnect (See Note 4)			900	264					900	264		
*Advantage Workstation for Signa CVMR Option					See Note 5						Not Applicable	
Existing Magnet Equipment Cabinets (ie. Magnet Power Supply, Shim Power Supply) as applicable to system												
Note * Optional equipment. † Maximum heat output is defined for temperature and humidity as defined in Table 4-1. 1 Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Blower Box intake which contains coil/patient cooling fans. 2 Shield Cooler Compressor Cabinet requires water cooling to dissipate heat output. Heat dissipation to air is negligible. 3 New Operator Workspace equipment includes the following: LCD Color Monitor, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, Keyboard, and interface modules mounted to Workspace Table 4 Values are for Main Disconnect Control option available from GEMS. Heat output may be different if customer supplied control is used. 5 Refer to Direction 2261309-100 Advantage Workstation 3.1 Ultra60 Pre-Installation.												

4-3-2 Water Cooling

Magnet equipment water cooling requirements are unchanged.

Gradient Coil

The Gradient Coil is air and water cooled. A water-to-air Water Chiller (WC1) is provided with the fixed site system. De-ionized water and anti-freeze mixture is supplied with the Gradient Coil for use in the closed loop system.

Note

The Water Chiller is a dedicated closed loop unit for Gradient Coil water cooling **only** and not to be shared with other equipment, refer to Section 4-3-3 Water Chiller (WC1) For Gradient Coil Cooling.

Note

The Gradient Coil has GE provided flexible hose (0.5 in. (12.7 mm) inside diameter) and adjustable compression clamps for connections. Refer to Section 6, INTERCONNECT DATA, for flexible hose routing information.

For Mobile systems (True Mobile, Transportable, Relocatable), the Gradient Coil water cooling will be provided by the Water-to-Water Heat Exchanger, the Van chilled water system will provide chilled water for the Heat Exchanger operation.

Shield/Cryo Cooler Compressor

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-3-2 Water Cooling** for Shield/Cryo Cooler Compressor water cooling requirements.

4-4 ALTITUDE

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-4 ALTITUDE** for additional information and specifications.

4-5 LIGHTING

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to **Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-5 LIGHTING** for additional information and specifications.

4-6 ACOUSTICS

Note

Design improvements in the MR system (RF/Gradient Body Coil, Magnet, RF and Gradient subsystems) balances the impact of the higher performance scan protocols such that there may be slight increases in the acoustics output. The following information is supplied for design of new installations but can also be used to assess the expected performance of the upgrade MR system.

The following acoustic information is provided for site planning and architectural design activities to address acoustics to meet local regulations and customer requirements. For more information about recommended safety procedures regarding patient exposure to MR-generated acoustic levels, see the system operator manual.

4-6-1 Background

A typical MR suite has two types of acoustic noise issues. The first is the acoustics of the room where patients and technicians are impacted by the noise of the MR system as the gradients are pulsed. The second is noise transmitted to other spaces via airborne and structure borne paths.

Airborne

The airborne transmission path entails the excitation of air within the magnet room; the resonator module consisting of the magnet, RF coil, and gradient coil generates acoustic noise similar to an intense loud speaker. The airborne noise passes through walls via any openings, i.e. small holes, cracks, HVAC ducts, and waveguides, into surrounding spaces within and possibly beyond the confinements of the building. Acoustic energy can transmit across distances of significant length.

Examples of airborne acoustic issues may include the following (not limited to only these):

- MR Operator exposure at Operator Workstation
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

Structureborne

The structureborne transmission path is the result of mechanical excitation of the floor/building structure causing the building to vibrate. The vibration of the surfaces at surrounding spaces then radiates as acoustic noise. Acoustic energy can transmit across distances of significant length.

Note

Less than 2 to 3% of installed base sites have experienced structureborne acoustic issues.

Examples of structureborne acoustic issues may include the following (not limited to only these):

- Areas directly above or below the Magnet Room, may not always be an issue
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

4-6-2 System Acoustic Noise Levels

Any GE factory-installed protocol can be modified by operators, which can increase or decrease acoustic SPL (Sound Pressure Level); or operators may create their own protocol which could produce a higher or lower acoustic SPL as stated under **Operating Conditions** Condition 1 below. Typical scans generate acoustic levels as stated under **Operating Conditions** Condition 2 below. In addition, the exposure times are completely under operator control. Consequently, hearing protection is required for all people in the Magnet Room during scans to prevent hearing impairment, acoustic levels may exceed 99dBA. Again, for more information about recommended safety procedures regarding patient exposure to MR-generated acoustic noise, see the MR system operator manual.

Ambient Conditions

To reduce any background noise due to cabinet blowers, etc., acoustical ceilings, walls, and floors are recommended. The following are typical noise level readings:

- Operator Area 55 dBA
- Equipment Room 75 dBA

Operating Conditions

Condition 1

MR scanners under “worst-case” operating conditions, could generate acoustic levels (as measured at the magnet iso-center) as follows:

Average SPL 118 dBA SPL= Sound Pressure Level
 Peak 128 dB
 Frequency Range 20 to 20k Hz

Condition 2

MR scanners for many typical clinical scanning scenarios though, generate acoustic levels (as measured at the magnet iso-center) somewhat lower as follows:

Average SPL 95 to 110 dBA
 Peak 110 to 120 dB
 Frequency Range 20 to 20k Hz

As recent history has shown an evolution towards more powerful (and hence louder) gradient subsystems, architects should consider the acoustic levels stated in the “worst case” Condition 1, mentioned above. Note that high-field Signa systems have the ability to run scanning protocols which can generate acoustic levels over the entire human perceptible frequency range (20 to 20k Hz), therefore attenuation over this entire range must be considered for site design.

4-6-3 Design Guidelines

Magnet Room

Noise generated by the MR system is inherent to the operation of the system, refer to Section 4-6-2 System Acoustic Noise Levels. The sound quality (human perception) within the Magnet Room can be modified by including sound absorbing materials to make the room sound more subdued and less harsh. The measured sound levels via a sound level meter will not change. However, the measured sound levels can be reduced only when the sound level generated by the MR System is reduced.

Sound quality improvements can be achieved by the following:

- Use ceiling tiles with fiberglass panels having a 2 inch (51mm) thickness set into the standard T-bar grid system.
- Adding fiberglass panels to the side walls covering approximately 20% of the side wall surface area. The panels should focus on covering the top half of the side walls. Panels could take many different and decorative shapes to improve the sterile look of the rooms. Typically panels might be on the order of 4ft x 6ft (1.2m x 1.8m) with a thickness of 4 inches (102mm) or equivalent. Panels shape could vary to produce mosaic effects to meet the customer preference. Any decorative materials used to cover the wall panels must be porous so that sound waves can pass through with ease. In principle, a person should be able to breath through the material with ease. Fire retardant cloth should be used. The NRC (Noise Reduction Coefficient) of the panels should be 0.95 or better when mounted against a hard surface such as drywall or concrete.

Inter-Spacial Areas

Acoustic noise control to mitigate noise from being transmitted to other spaces often amounts to paying attention to small details while working with ordinary construction materials. The key objectives are to eliminate all cracks and gaps in the wall construction while making sure that the doors, walls, floor, and ceiling have adequate transmission loss via mass or special double wall construction along with good fitting massive doors.

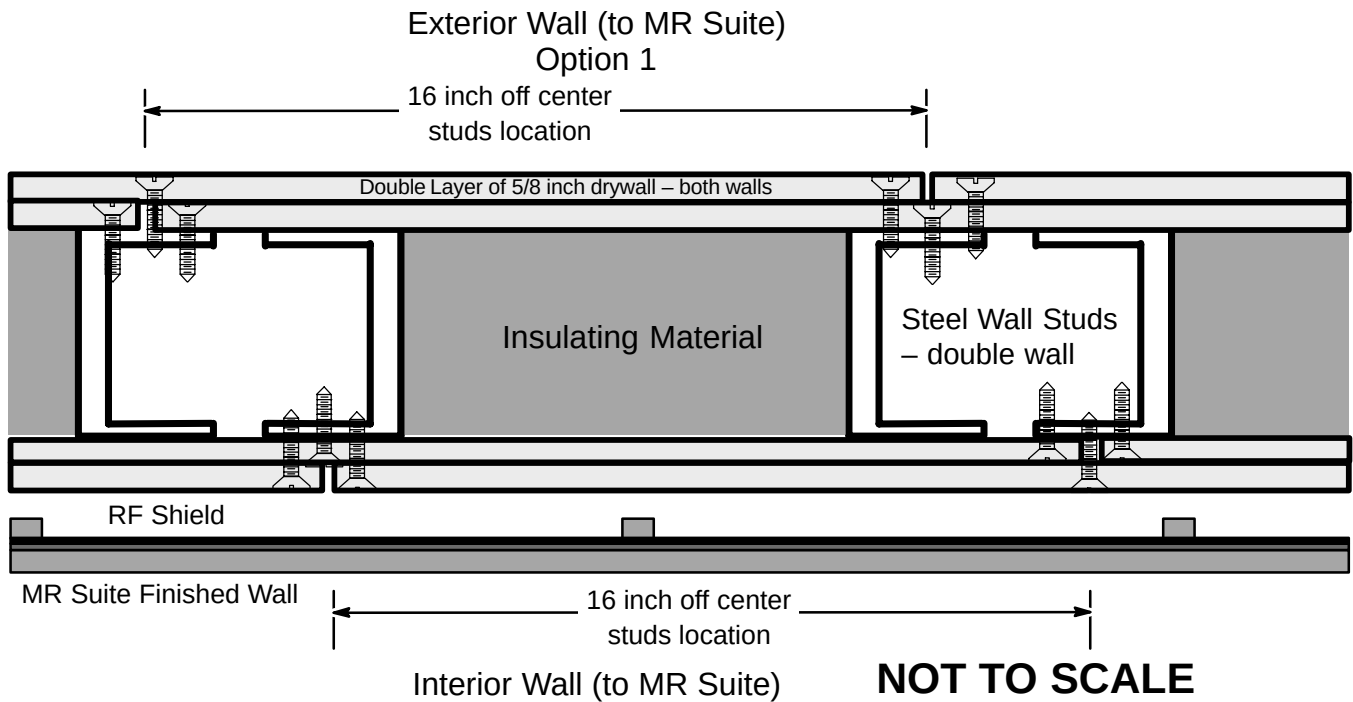
The entire Magnet must be surrounded by walls with substantial mass and/or double wall construction so that noise is contained in the room and not allowed to pass through into nearby spaces. Wall junctions must be sealed with acoustical sealant so that noise waves do not escape from the room. In principle, if the room were filled with smoke and under a positive pressure, no smoke would leak from the room.

Wall Construction

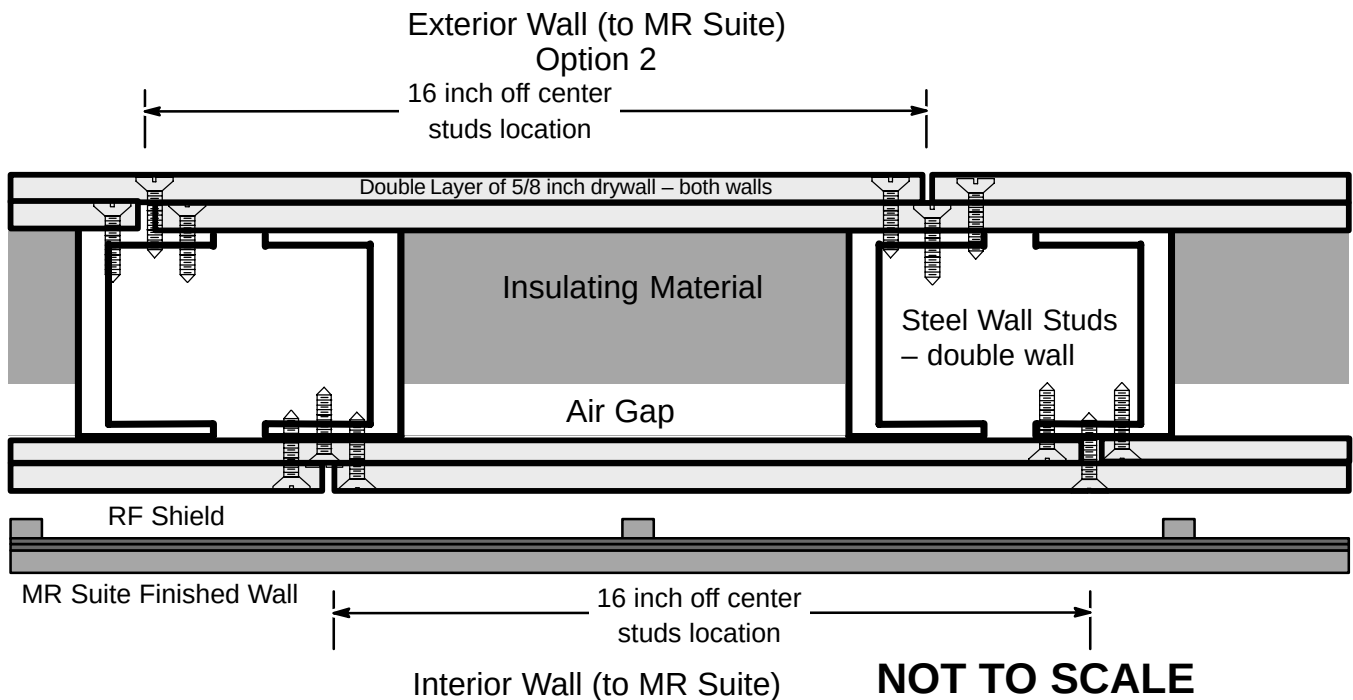
Wall Construction will entail ordinary building materials in a careful configuration.

- The preferred wall would have an ASTM STC 50 construction which entails the use of standard wall construction of steel studs (typically 3-5/8 inch (92 mm)) with 2 layers of Type X drywall (typically 5/8 (16 mm)) on each side totaling 4 layers and fiberglass batt in the stud cavity. All drywall must be overlapped by 6 inches (152 mm) or more. Beads of (USG) Acoustical Caulking (non-hardening) would be used around the entire perimeter of the drywall. Any form of wall penetration should be avoided. Any necessary wall penetrations must be sealed using combination of Acoustical Caulking (non-hardening) and fiberglass batt material. See examples of wall construction shown in Illustrations 4-1 and 4-2.
- The top of the wall must join the ceiling/floor above so that no cracks or gaps occur. If metal pan is used on the ceiling/floor (above), then flute seals would be used to seal the gaps between the drywall and the pan. Alternately drywall can be cut out to fit into the flutes. Acoustical caulking (non-hardening) will be used to seal the remaining cracks and gaps.

4-6-3 Design Guidelines (Continued)



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL - OPTION 1
ILLUSTRATION 4-1



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL - OPTION 2
ILLUSTRATION 4-2

4-6-3 Design Guidelines (Continued)High Bay RF Room

A high bay RF Room is a self contained RF Room which has open air space between the RF Room ceiling and the building floor above. The air space is an acoustic transmission path. Acoustic energy must be reduced to minimize this transmission of energy through this path.

In cases where the Magnet is to be installed in a high bay, it may be most effective to enclose the RF Room with its own drywall and steel stud room. The key difference being a ceiling assembly that mimics the sidewall construction to contain noise.

- Normal high STC stud walls from above would be used to support a ceiling assembly constructed of structural C channel with two layers of drywall on each side (total of 4 layers) with fiberglass batt in the cavity.
- Penetrations should be avoided via the use of surface mounted lights. HVAC and ducts passing through the ceiling, party wall or side walls would require acoustic noise attenuation in the form of inline silencers. Gaps and cracks would be sealed between the ceiling, party wall or vertical side walls and the cryogen vent plumbing. In essence the Magnet would be enclosed in a drywall "doghouse".

Miscellaneous Plumbing, RF Windows and RF Doors

Other construction details are equally important to mitigate noise transmission to meet the intended goal.

- Pipes (gas or water) and electrical conduit or Magnet Room signal cables must be sealed where they penetrate the walls or ceiling. A heavy mastic material such as Duxseal™ is appropriate.
- RF windows should be purchased as window/frame units with an STC rating obtained from laboratory testing per ASTM standards. STC 50 to 60 windows are needed. The installation must include proper sealing to avoid sound leaks.
- RF doors should be selected to provide and STC 50 to 50 to quell the noise. Contact RF Shield Room supplier for selection of RF doors that meet the local acoustic codes and site acoustic requirements. RF door seals must be selected to prevent small gaps around the door perimeter and at the door threshold. RF door seals would either require periodic replacement or a door seal that would last the life of the Magnet Room.

4-7 ROOM VENTILATION

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation Section 4-7 ROOM VENTILATION*** for requirements and recommendations.

4-8 CRYOGENIC VENTING


WARNING!

IF AN UPGRADE TO AN LCC MAGNET IS BEING PERFORMED WITH THE SIGNA SYSTEM UPGRADE THEN REFER TO *DIRECTION 2223170 SIGNA MR/i & CV/i PRE-INSTALLATION SECTION 4-8 CRYOGENIC VENTING* FOR REQUIREMENTS AND RECOMMENDATIONS FOR LCC MAGNET CRYOGENIC VENTING.

4-9 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS

If an upgrade to an LCC magnet is being performed with the Signa system upgrade then refer to ***Direction 2223170 Signa MR/i & CV/i Pre-Installation SECTION 4-12 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS*** for requirements and recommendations..

4-10 CONSTRUCTION MATERIALS

Systems installing an upgrade to LCC Magnet must meet the requirements in ***Direction 2223170 Signa MR/i & CV/i Pre-Installation SECTION 4-13 CONSTRUCTION MATERIALS*** for requirements and recommendations..

4-11 VIBRATION

4-11-1 Definition

Improvements in MR imaging technology has increased imaging capabilities. Certain MR procedures require a stable environment to achieve high resolution image quality. **Environmental site vibration has the ability to affect the magnet phase stability and ultimately image quality.** The vibration effects on image quality can be minimized early in the site planning for Signa Horizon upgrades.

The magnet may be sensitive to vibration in the frequency range of 0.5 to 45 Hz depending on the amplitude of the vibration. In the physical area where the MR system is located, every precaution must be taken to ensure that the vibration is minimized. In the magnet siting area, the structural stability and behavioral characteristics can be assessed. **The stability tests (HSS) currently used by local GE Field Service to evaluate the existing system performance and image quality can be used to assess the vibration environment. Sites which currently pass their system stability criteria (HSS) may proceed with the upgrade. Sites which have marginal stability performance require vibration measurements to be performed to identify vibration source(s).** Then it is the customer's responsibility to contract a vibration consultant to perform vibration measurements as defined in Appendix A, MR SITE VIBRATION TEST GUIDELINES. With the vibration consultant, local GE Field Service and/or Installation Specialist must verify the elimination/reduction of all identified sources do improve the vibration environment. GE can assist in interpreting marginal site test results and predicting the impact on system performance. However it is ultimately the customer/architect/engineer responsibility to design a site solution.

4-11-2 Types Of Vibration Image Quality Issues

Vibration capable of affecting MR image quality can be classified as either steady state or transient vibration. Steady state vibration typically refers to disturbances caused by rotating machinery. Examples of machinery known to have previously generated vibration image quality problems are exhaust fans, air conditioning blower units, compressors, pumps (air and water), etc. Steady state vibration would be measurable during long time intervals. Transient vibrations are typically a function of the building structure or the building foundation. Transient vibration would typically be associated with vehicular traffic, pedestrian motion, patient transport, door slamming, etc. Transient vibration would typically occur only during short periods of time. The transient event would typically decay from a high vibration amplitude to lower levels in short periods of time.

Steady State Vibration Issues

1. Power down all sources exceeding vibration guidelines.
2. Run MR system stability test.
 - a. If all sources have been properly identified the MR system will meet GE specifications.
 - b. If GE specifications are exceeded, continue searching for additional vibration sources.

4-11-2 Types Of Vibration Image Quality Issues (Continued)

Transient Vibration Issues

Vibration measurements and system stability tests must be performed during the same period of time. Only when the vibration event(s) and phase stability deviation(s) occur at the same time will transient vibration be confirmed as source of system stability.

1. Measure the MR system stability and vibration, simultaneously, during normal system scanning hours. Transient vibration and transient phase stability deviation exceedance(s) must occur at the exact same time for transient vibration to be an imaging concern.
2. If transient vibration produces imaging concerns, the vibration consultant must identify the source(s) of the transient in which case additional testing is needed. Schedule and perform additional vibration testing during periods of time when the suspected transient vibration issues can be controlled, limited, or eliminated. Evaluate the MR system performance during periods of time per the following.
 - a. If all sources have been properly identified the MR system will meet GE specifications.
 - b. If GE specifications are exceeded, continue searching for additional vibration sources.

Magnet Siting Requirement

Improper mechanical coupling of the magnet to the floor can in some circumstances create unique vibration characteristics. Magnets improperly mechanically coupled to the floor may experience steady state and/or transient vibration system stability issues. When the system stability tests do not meet acceptance criteria the vibration testing defined in Appendix A will help to identify if the magnet is properly mounted to the floor. The customer is not limited to the GE options for mounting the magnet: site conditions may necessitate alternative methods.

4-11-3 Specifications For 1.5T S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed **10⁻⁵ g rms** maximum single frequency **above ambient baseline** from **0.5 to 45 Hz**. In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.

Transient Vibration

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** level defined above.

4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed the following maximum single frequency components **above ambient baseline**:

- 1×10^{-5} g rms 0–15 Hz
- 5×10^{-5} g rms 15–20 Hz
- 1×10^{-5} g rms 20–35 Hz
- 2×10^{-4} g rms 35–45 Hz

In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.

Transient Vibration

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** levels defined above.

4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed the following maximum single frequency components **above ambient baseline**:

- 1×10^{-5} g rms 0–10 Hz
- 5×10^{-5} g rms 10–30 Hz
- 1×10^{-5} g rms 30–38 Hz
- 2×10^{-4} g rms 38–45 Hz

In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.

Transient Vibration

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** levels defined above.

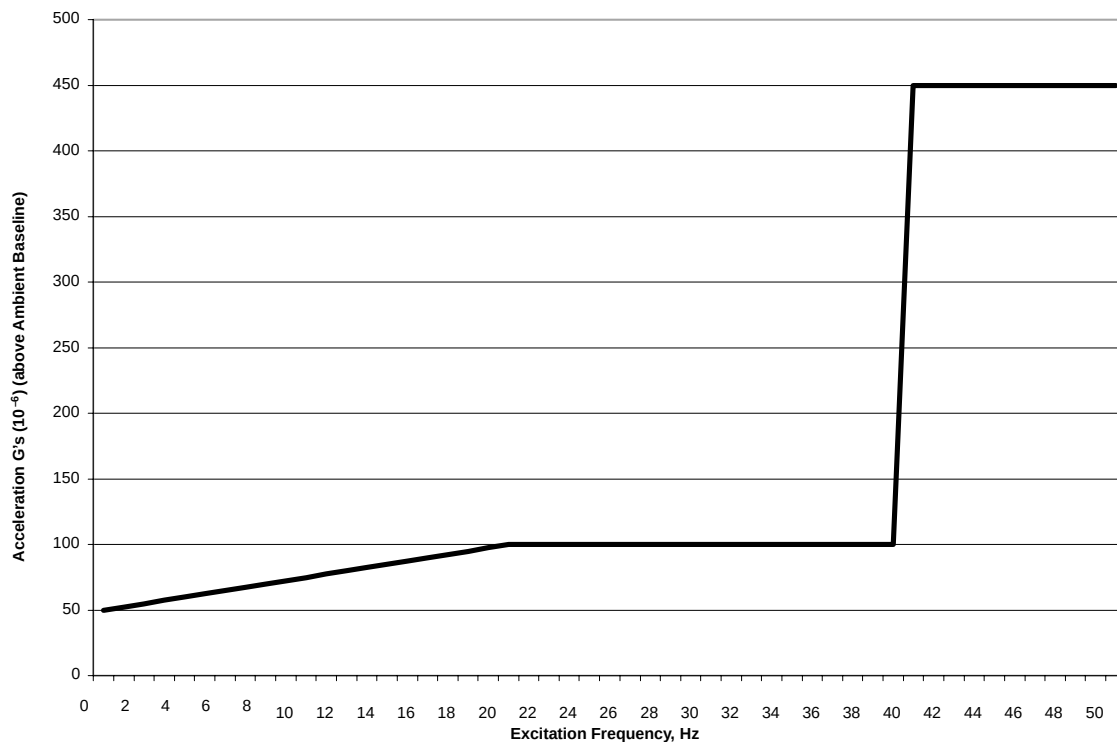
4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed the following maximum single frequency components **above ambient baseline** (refer to Illustration 4-3):

- 5×10^{-5} g rms at 0 Hz ramping to 10×10^{-5} g rms at 20 Hz
- 10×10^{-5} g rms 20-40 Hz
- 45×10^{-5} g rms 40-50 Hz

In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.



LCC MAGNET WITH GRADIENT ISOLATION STEADY STATE VIBRATION SPECIFICATIONS

ILLUSTRATION 4-3

Transient Vibration

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** levels defined above.